## DTC P253F:00 [SKYACTIV-D 1.5]

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| DTC P253F:00 | Engine oil deteriorated |
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| DETECTION CONDITION | • If any of the following conditions are met:<br><br>**Note**<br>• The PCM determines that the engine oil has deteriorated due to an engine oil pressure decrease. Dilution of the oil from fuel becoming mixed in can be considered a cause of the engine oil pressure decrease.<br>— The amount of engine oil dilution exceeds the specified value during the diesel particulate filter regeneration control. (European (L.H.D./U.K.))<br>— The amount of engine oil dilution exceeds the specified value (fluctuates by ambient temperature).<br>— With the following driving conditions met, the engine oil pressure decreases **51 kPa {0.52 kgf/cm$^2$, 7.4 psi} or more** compared to when the engine oil was replaced:<br>**MONITORING CONDITIONS**<br>• Engine speed: **1,275—1,325 rpm**<br>• Engine oil temperature: **89.5—90.5 °C {193.1—194.9 °F}**<br>• Traveled distance after engine oil replacement is specified value* or more<br>*: Specified value:<br>— **12,000 km {7,456.5 mile} or more** (European (L.H.D.)/Israel)<br>— **7,000 km {4,350 mile} or more** (except for European (L.H.D.)/Israel)<br>**Diagnostic support note**<br>• This is an intermittent monitor (other).<br>• The check engine light does not illuminate.<br>• FREEZE FRAME DATA/Snapshot data is not available.<br>• DTC is not stored in the PCM memory. |
| FAIL-SAFE FUNCTION | Not applicable |
| POSSIBLE CAUSE | • Engine oil pressure sensor malfunction<br>• Engine oil temperature sensor malfunction<br>• Engine oil attenuation by frequently diesel particulate filter regeneration<br>• Engine oil data reset procedure does not perform in each engine oil replacements.<br>• Oil viscosity lowered due to engine oil deterioration<br>• PCM malfunction |
| SYSTEM WIRING DIAGRAM | Not applicable |

### Diagnostic Procedure

| STEP | INSPECTION | | ACTION |
|---|---|---|---|
| 1 | **VERIFY RELATED SERVICE INFORMATION AVAILABILITY**<br>• Verify related Service Information availability.<br>• Is any related Service Information available? | Yes | Perform repair or diagnosis according to the available Service Information.<br>• If the vehicle is not repaired, go to the next step. |
| | | No | Go to the next step. |
| 2 | **VERIFY RELATED PENDING CODE AND/OR DTC**<br>• Switch the ignition off, then ON (engine off).<br>• Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)<br>• Are any other PENDING CODEs and/or DTCs present? | Yes | Go to the applicable PENDING CODE or DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].) |
| | | No | Go to the next step. |
| 3 | **INSPECT ENGINE OIL PRESSURE SENSOR**<br>• Inspect the engine oil pressure sensor. (See ENGINE OIL PRESSURE SENSOR INSPECTION [SKYACTIV-D 1.5].)<br>• Is there any malfunction? | Yes | Replace the engine oil temperature sensor/engine oil pressure sensor, then go to Step 6. (See ENGINE OIL TEMPERATURE SENSOR/ENGINE OIL PRESSURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) |
| | | No | Go to the next step. |

| STEP | INSPECTION | | ACTION |
|---|---|---|---|
| 4 | **INSPECT ENGINE OIL TEMPERATURE SENSOR**<br>• Inspect the engine oil temperature sensor.<br>(See ENGINE OIL TEMPERATURE SENSOR INSPECTION [SKYACTIV-D 1.5].)<br>• Is there any malfunction? | Yes | Replace the engine oil temperature sensor/engine oil pressure sensor, then go to Step 6.<br>(See ENGINE OIL TEMPERATURE SENSOR/ENGINE OIL PRESSURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) |
| | | No | Go to the next step. |
| 5 | **REPLACE ENGINE OIL**<br>• Replace the specified engine oil.<br>(See ENGINE OIL REPLACEMENT [SKYACTIV-D 1.5].) | — | Perform the “ENGINE OIL DATA RESET”, then go to the next step.<br>(See ENGINE OIL DATA RESET [SKYACTIV-D 1.5].) |
| 6 | **VERIFY DTC TROUBLESHOOTING COMPLETED**<br>• Clear the DTC from the PCM memory using the M-MDS.<br>(See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].)<br>• Start the engine and warm it up completely.<br>**Caution**<br>**• While performing this step, always operate the vehicle in a safe and lawful manner.**<br>**• When the M-MDS is used to observe monitor system status while driving, be sure to have another technician with you, or record the data in the M-MDS using the PID/DATA MONITOR AND RECORD capturing function and inspect later.**<br>• Drive the vehicle under the FREEZE FRAME DATA/snapshot data condition.<br>• Perform the DTC Reading Procedure.<br>(See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)<br>• Is the same DTC present? | Yes | Repeat the inspection from Step 1.<br>• If the malfunction recurs, replace the PCM.<br>(See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)<br>Go to the next step. |
| | | No | Go to the next step. |
| 7 | **VERIFY AFTER REPAIR PROCEDURE**<br>• Perform the “AFTER REPAIR PROCEDURE”.<br>(See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].)<br>• Are any DTCs present? | Yes | Go to the applicable DTC inspection.<br>(See DTC TABLE [SKYACTIV-D 1.5].) |
| | | No | DTC troubleshooting completed. |